Test 1 review day

CMSE 381

Prof. Mengsen Zhang

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Mon, Feb 10, 2025

Announcements

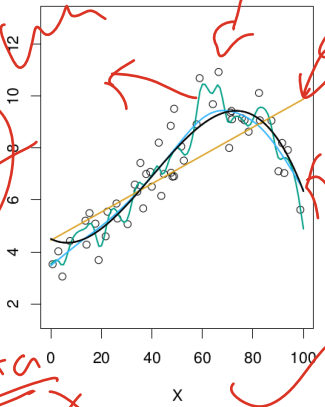
Announcements:

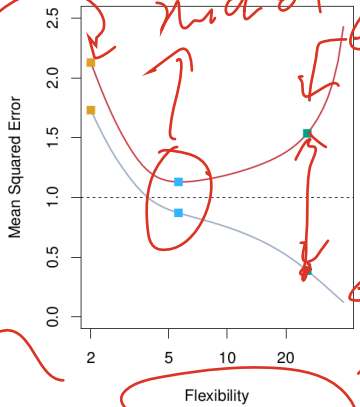
- midterm # 1 on Wednesday
- Project info, honors option posted on course website

Based on the questions submitted

- Bias Variance Trade off
- KNN classification
- How to prepare

Training error vs test error





Train

Test error

Test data

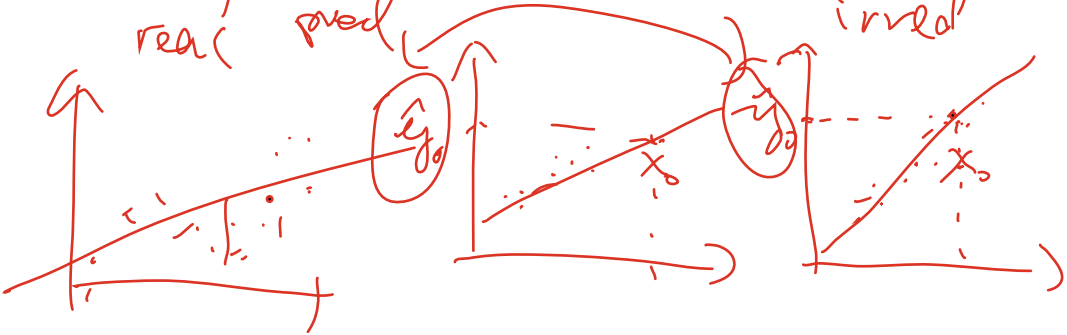
test error

Bias-variance

test

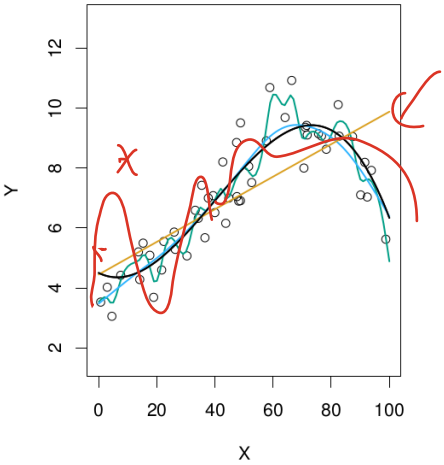
$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

real pred irred



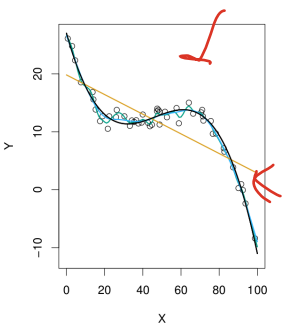
Variance

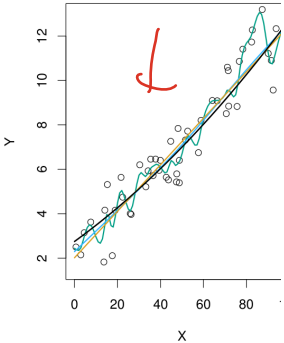
Variance: the amount by which $\hat{f}$ would change if we estimated it using a different training data set.



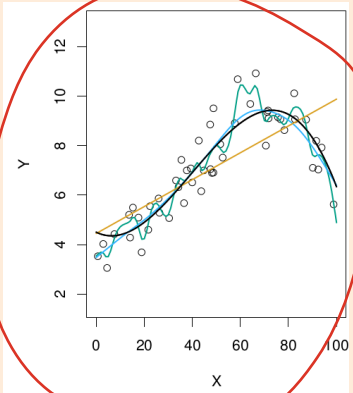
Bias

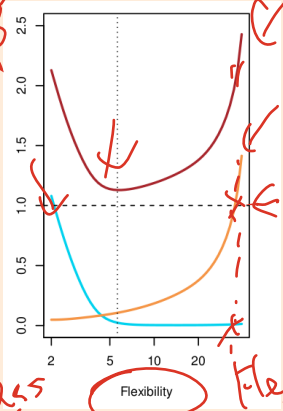
Bias: the error that is introduced by approximating a (complicated) real-life problem by a much simpler model.





Which is which





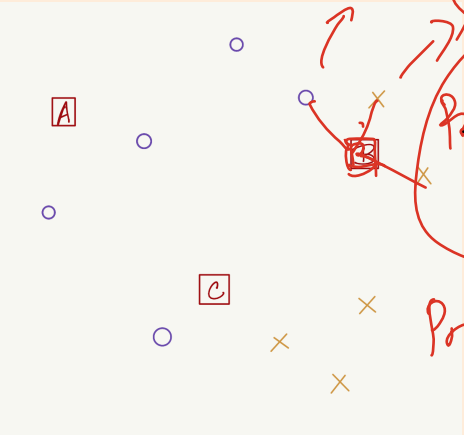
Label the line corresponding to each of the following:

- MSE
- Bias
- Variance of $\hat{f}(x_0)$
- Variance of ε

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

Review KNN classifier example

Here label is shown by O vs X. What are the knn predictions for points A, B and C for $k = 1$ or $k = 3$?



$\Pr(B \to X \mid \text{data}) = \frac{2}{3}$

$\Pr(B \to O \mid \text{data}) = \frac{1}{3}$

Point	$k = 1$ Prediction	$k = 3$ Prediction
A		
B		X
C		

Preparing for exam

- review slides
- write down key concepts equations on the cheatsheet, annotate them
- find pictures, equations, results tables you don't understand. bring to this class or read up in text book.

a bring calculator